

Q1. What is a Data Structure? Give 3 real-life examples.

Data Structure is a method of organizing and storing data efficiently for easy access and modification.

Examples: Stack of plates, Queue at a ticket counter, Folder hierarchy in a computer.

Q2. Differentiate between Linear and Non-linear Data Structures.

Linear: Elements are arranged sequentially (Array, Stack, Queue, Linked List).

Non-linear: Elements are connected hierarchically (Tree, Graph).

Q3. Explain the internal structure of an Array with a memory diagram.

An array stores same-type elements in contiguous memory.

Diagram:

Index: 0 1 2 3

Value: 10 20 30 40

Addr :1000 1004 1008 1012

Q4. Write the algorithm and C program for Array insertion.

Algorithm: Read array, read position and item, shift elements to the right, insert item, display updated array.

Q5. Explain Stack using LIFO concept with a diagram.

Stack follows LIFO (Last In First Out).

Diagram:

Top

[30]

[20]

[10]

Pop removes 30 first.

Example: Stack of plates.

Q6. What is the difference between Stack and Queue?

Stack: LIFO, one end.

Queue: FIFO, insertion at rear and deletion from front.

Q7. Write C code for Enqueue and Dequeue operations.

Enqueue inserts an element at the rear. Dequeue removes an element from the front. Both operations are $O(1)$ in a simple queue.

Q8. Explain Singly Linked List with a node diagram.

Each node contains Data and Next pointer.

Diagram:

Head -> [10|*] -> [20|*] -> [30|NULL]

Example: Music playlist.